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MAGNETISM & ELECTROMAGNETISM

For Students, Professionals and Beyond

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1. Introduction to Magnetism

Magnets and magnetism play an important role in Electrical and Electronic Engineering because without them electrical components such as relays, solenoids, inductors, chokes, transformers, loudspeakers, motors, generators and a host of other electro-magnetic devices would not work, if **magnetism** did not exist.

Magnetism and electricity are closely related, since thanks to magnetism, whether natural or artificial (electromagnetism), a lot of electrical energy can be produced. Magnetism is a very natural but invisible force which can be used, created and controlled to produce most of the world's electrical power. That's because when used by an electrical generator, magnetism can create electrical energy. Then a magnetic field can transport energy.

Generally, magnets are any ferromagnetic material which can be used to attract iron or material alloys which contain iron. All magnets contain two distinct magnetic poles that exert a force of attraction or repulsion on each other and as such are surrounded by a magnetic field which has both direction and strength.

Not all metals however are suitable for use as magnetic materials. For example, aluminium cannot be magnetised. Only those classed as ferromagnetic materials such as iron, nickel, or cobalt can be magnetised because their magnetic domains can be easily aligned.

This means that when a metallic material becomes magnetised, the arrangement of the atoms inside the material become "lined-up" and can stay perfectly aligned for a very long time producing what is called a permanent magnet similar to that shown in Figure 1.

But before we look at **Magnetism** and especially **Electromagnetism** in more detail, we need to remember back to our physics classes of how magnets and magnetism works.

Figure 1. A Typical Horseshoe Permanant Magnet

There are basically two types of magnets, "Permanent Magnets" and "Temporary Magnets". A permanent magnet is simply a piece of magnetised iron or steel which has the ability to attract towards itself other pieces of iron or steel (as well as a few other types of ferrous materials).

Permanent magnets are magnetic materials that do not require any other external force or power to keep their magnetic field strong and active, and as such are capable of retaining their ability to attract other metals for many years as long as their atomic domains remain aligned.

However, as their name suggests, temporary magnets are those whose magnetic field can be turned ON or OFF at any time, and therefore controlled. As such they have no ability to retain their magnetised state.

One common way of controlling a magnetic field is by using electricity. If an electrical current flows through a wire or conductor, it will create a magnetic field around itself. Thus there exists a direct link between the use of electricity to create magnetism. The production of magnetism using electricity commonly called "*electromagnetism*".

> *Magnetism is the force of attraction or repulsion between any two magnetic poles of north and south*

As we will see later, electromagnets depend on the flow and amount of electrical current to both produce and to keep their magnetic field active with the strength of the magnetic field produced proportional to the amount of current flow. Then every coil, loop or length of wire can use the effect of electromagnetism to their advantage in a wide variety of applications.

2. The Nature of Magnetism

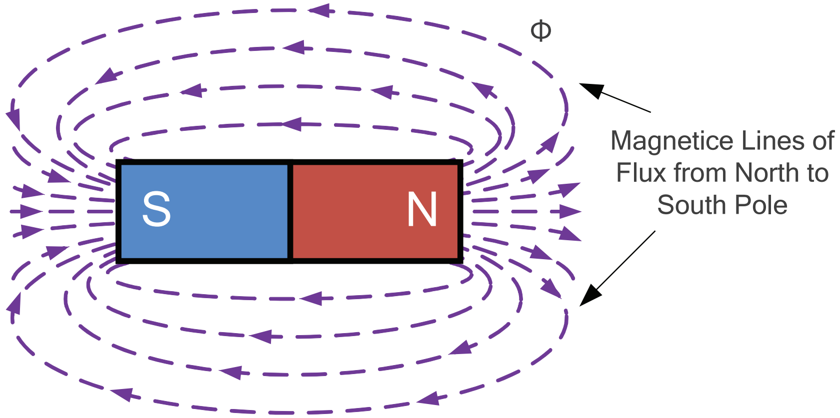
Permanent magnets can be found in their natural state in the form of a magnetic ore, with the two main types being Magnetite and Lodestone. Both these types of magnetic materials produce a 3-dimensional magnetic field around themselves which consists of lines of magnetic flux.

All magnets, no matter what their shape, have two distinct regions or ends called magnetic poles and are surrounded by a magnetic field. The shape of this magnetic field is more intense in some parts around the magnet than others.

The area around the magnet which has the greatest concentration of magnetic lines of flux being around the point where the magnet has a pole.

This is because the lines of flux are closer together around the poles resulting in a stronger magnetic force. The magnetic poles are located inside the magnet as shown in Figure 2.

Figure 2. Lines of Flux Around a Permanent Magnet



Magnetic fields are always shown visually as lines of force that give a definite pole at each end of the material where the flux lines are much denser and concentrated. The lines which go to make up a magnetic field showing the direction and intensity are called Lines of Force or more commonly "Magnetic Flux", and are given the Greek symbol, Phi (Φ) as shown in Figure 2.

These lines of flux (called a vector field) cannot usually be seen by the naked eye, but they can be seen visually by using iron fillings sprinkled onto a sheet of paper with the magnet below, or by using a small compass to trace them out. This will show the region around the magnetic in which the magnetic flux can be detected.

There is always a region or end of a magnet called the North-pole (N) and there is always an opposite region called the South-pole (S). The "direction" of the magnetic field, at any point along it is always from north-to-south. Thus magnetic poles are always together in pairs (one north, one south) and have polarity.

However, a magnets magnetic flux does not actually flow or move from the north to the south pole or flow anywhere for that matter as magnetic flux is a static region around a magnet in which the magnetic force exists. In other words, magnetic flux does not flow or move it is just there and is not influenced by gravity.

Some important facts you will notice when plotting magnetic lines of flux. These are:

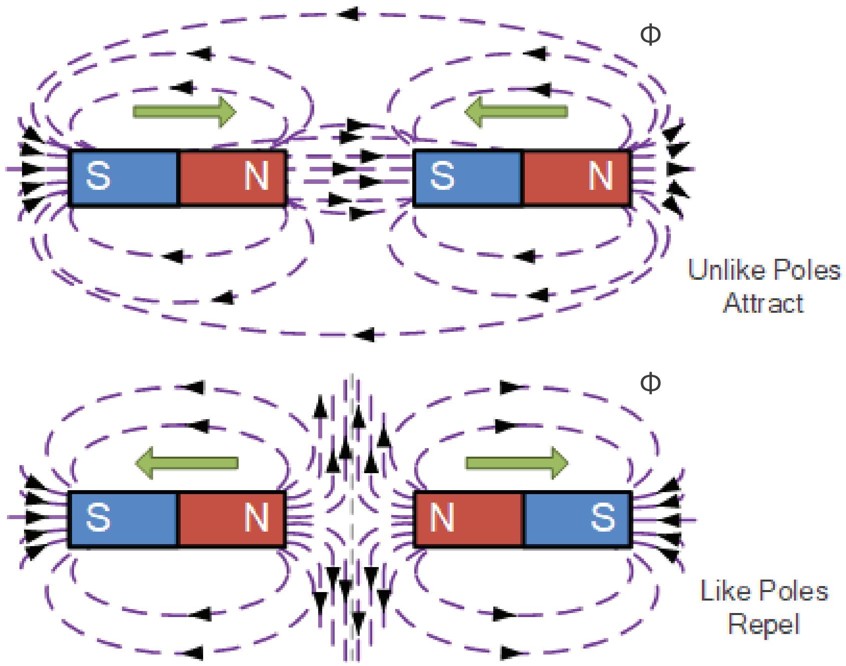
- Magnetic lines of flux NEVER cross each other.
- Magnetic lines of flux are CONTINUOUS from pole to pole.
- Magnetic lines of flux always form individual CLOSED LOOPS around the magnet.
- Magnetic lines of flux have a definite DIRECTION from North to South.
- Magnetic lines of flux that are close together indicate a STRONGER magnetic field.
- Magnetic lines of flux that are farther apart indicate a WEAKER magnetic field.

Magnetic lines of flux can either attract and repel each other. When two permanent magnets are brought close together, the interaction between the two magnetic poles causes one of two things to occur as the magnets exert a magnetic force upon each other:

1. When the two magnetic poles are exactly the same, (north-north or south-south) they will REPEL each other.
2. When the two magnetic poles are not the same, (north-south or south-north) they will ATTRACT each other.

This effect is easily remembered by the expression that "opposites attract" as opposite magnetic poles attract each other. The effect upon the magnetic fields due to the force of the poles can be shown in Figure 3. as like poles repel and unlike poles attract.

Figure 3. Interaction of Magnetic Fields with Like and Unlike Poles



The interaction of two magnetic fields shows the lines of force around the magnet. As we said previously, magnetic poles always occur in pairs within the same magnet. So if we split, cut or break a permanent magnet into two or more pieces, then each piece will have its own north and south pole as each piece gains a new pole.

Thus magnetic poles always exist in pairs consisting of one north (N) and one south (S) pole which cannot be isolated. There is no single Mono-pole.

3. Magnetic Flux Density (B)

The lines of magnetic flux around a magnetic material is given the Greek symbol, Phi, (**Φ**) with the unit of flux being the Weber, (**Wb**). The number of lines of force within a given unit area is called the "Flux Density" and since flux (Φ) is measured in (Wb) and area (A) in metres squared, (m^2).

Flux density is therefore measured in Webers per Metre^2 of the number of magnetic flux lines per square unit of area. The standard unit of magnetic flux density is commonly given as Wb/m^2 or tesla [T] (no, not after the car) after Nikola Tesla. Therefore we can say that, one Wb/m^2 is equal to one Tesla, $1Wb/m^2 = 1T$. This unit is given the symbol "**B**" and is a measure of flux concentration in each square metre of magnetic field.

Flux density (B) is proportional to the lines of force and inversely proportional to area so we can define magnetic flux density as being:

$$\text{Magnetic Flux Density, } B = \frac{\Phi}{A} \ Wb/m^2$$

Note that all calculations for flux density are done in the same units, e.g. flux (**Φ**) in Webers, area (**A**) in m^2 and magnetic flux density (**B**) in Teslas producing a vector quantity.

Clearly then, magnetic flux density will be greater around the magnetic poles of a magnet because the density of the field lines is greater per given area.

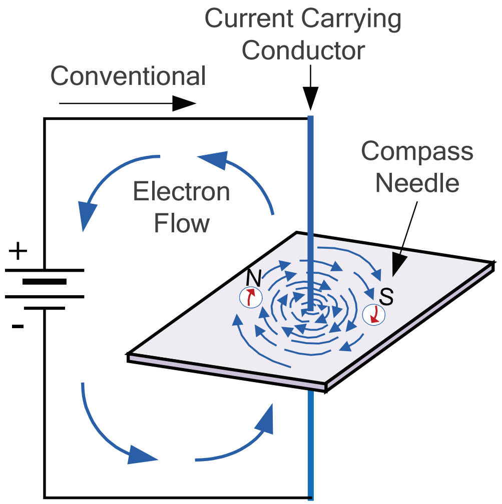
4. Electromagnets and Solenoids

The magnetism and magnetic fields produced by the effect of electricity flowing through a conductor is called **Electromagnetism**. The generation of magnetism finds numerous applications in electrical engineering such as electrical generators, motors, transformers, and relays, etc. Thus electromagnets are basically temporary magnets whose magnetic field is created and controlled by an electric current, either AC or DC.

Electromagnetism is based on the fact that when an electric current flows through a length of wire (a conductor), it creates a circular magnetic field around itself as shown in Figure 4.

The strength and direction of rotation of this magnetic field around the wire is determined by the intensity and direction of the electric current flowing through the conductor from positive to negative. Then every length of wire has the ability to produce the effect of electromagnetism around itself when an electrical current flows through it.

Figure 4. The Magnetic Field Around a Conductor



The corresponding magnetic field produced will be stronger near to the centre of the current carrying conductor because the path lengths of the magnetic loops are greater the further away from the conductor resulting in weaker flux lines.

The issue here is that the magnetic field generated around a straight length of current-carrying wire is very weak even with a high current passing through it.

However, if we wound the wire into several loops together along the same axis producing a coil of wire, the resultant magnetic field will become much more concentrated and stronger than that of just one single loop. This then produces an electromagnetic coil more commonly called a Solenoid.

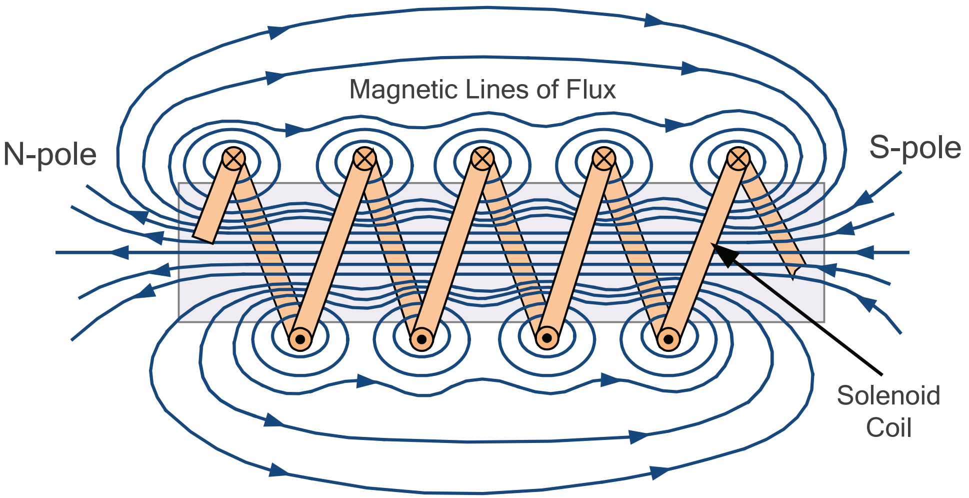
5. The Solenoid

A solenoid is simply a series of wire loops wound tightly to create a single coil. Solenoids behave like bar magnets with a distinct north and south pole when an electrical current passes through the coil.

The static magnetic field produced by each individual coil loop is summed with its neighbouring magnetic loop. Thus the combined magnetic field is more concentrated in the air-cored centre of the coil.

Then the resultant static magnetic field with a north pole at one end and a south pole at the other is uniform and much more stronger within the centre of the coil than around the exterior as shown in Figure 5.

Figure 5. Magnetic Field Around a Solenoid



The magnetic field that is produced by a long solenoid is stretched out in a form of a permanent magnet, producing a distinctive North (N) and South (S) pole. The amount and strength of the lines of flux being proportional to the amount of electric current flowing through the coil windings.

If additional loops of wire are added and wound upon the same coil with the same current flowing, the magnetic field strength will be increased. Then the amount of magnetic flux available in any given electromagnetic circuit is directly proportional to the current flowing through it and the number of turns of wire within the coil.

The product of the number of coil turns, N and the current, I flowing through it is called *Magneto-motive Force*, (m.m.f). It is this m.m.f which is creating the magnetic field of the coil. Therefore, m.m.f = N x I given in Ampere-turns (AT).

We can therefore see that the magnetic field strength of an electromagnet is determined by the ampere turns of the coil with the more turns of wire in the coil the greater will be the strength of the magnetic field generated by it.

The magnetic field strength of "magnetising force" created by an air-cored long solenoid, with a fixed m.m.f and length, l (in metres), is assumed to be constant value. Then:

$$\text{Magnetising Force, } H = \frac{\text{m.m.f}}{l} = \frac{N \times I}{l} \ \text{A/m}$$

Then a current, (i) flowing through the solenoid will produce a uniform magnetic field strength, H expressed in Amperes per metre (or Ampere-turns per metre, AT/m). Thus the magnetic field intensity of a 60-turn, 12cm long solenoid coil, with 3 amperes flowing in it would be: 1500 A/m.

But this magnetising force creates a magnetic field of Φ Webers, and as we have seen previously, the magnetic flux density in webers per square metre, Wb/m^2 or Tesla's (T) is given the symbol B.

$$\text{Flux Density, } B = \frac{\Phi}{A} = \frac{\mu_o N I}{A} = \mu_o H$$

Where: A is the cross-sectional area of the solenoid coil in square-metres, m^2, and is the permeability of the magnetic field in free-space (air-cored). The permeability of free space is the ratio of B to H.

6. Permeability of Magnetic Materials

Then we can see that for a given electromagnetic coil, its field strength and intensity depends greatly on the amount of current (in Amperes) flowing through its coil as well as the number of coil turns within a given length.

But we can increase and strengthen the magnetic field created around a wound coil even further from that of an air-cored solenoid by winding the coil turns around a soft iron core or another ferromagnetic type material which is a good conductor of magnetic flux.

Permeability is the ability or ease by which a material accepts (or resists) the flow of magnetic lines of flux through itself. So a material with high permeability, iron or steel for example would offer less resistance to flux lines compared to air or a vacuum and can therefore be easily magnetised and vice-versa. Then the use of an inner core with a high value of permeability would produce large amounts of flux with only a few turns of wire.

Note that air or a vacuum is the poorest conductor of magnetic flux and as such has a very low value of permeability. The numerical constant for the permeability of free space is given as: $\mu_o = 4\pi \times 10^{-7}$ Henries per metre, (H/m).

Permeability is the measure of how easily magnetic flux can pass through a material compared to air

If different materials with the same physical dimensions are used as the core for an electromagnet, the strength of the magnetic flux will vary in relation to the core material being used.

The measure or comparison of a particular materials permeability compared with air (commonly a vacuum) is called relative permeability. The symbol for relative permeability is μ_r (Greek letter mu), where the subscript "r" stands for relative.

Thus absolute permeability, μ, is defined as being:

$$\mu = \mu_o \mu_r \ \text{H/m}$$

Where:

$$\mu_r = \frac{\mu}{\mu_o}$$, the relative permeability of the core

Thus, relative permeability is the product of μ (absolute permeability) and μ_o the permeability of free space allowing core materials to be either magnetic or nonmagnetic characteristics. So we can correctly say that the magnetic flux density around a coil with a solid core is given as: B = H (teslas).

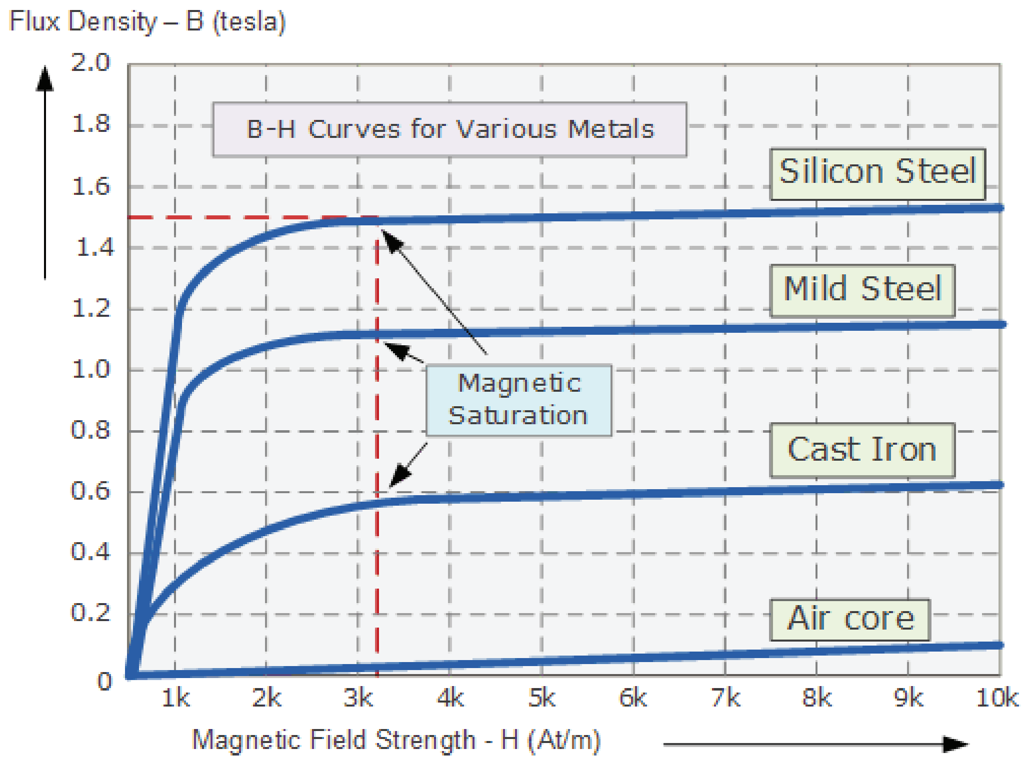
So if a 350 AT/m electromagnet is wound on a hard steel core that has a relative permeability of 1600, the magnetic flux density produced by the coil would therefore be given as: $B = \mu H = 1600 \times 4\pi \times 10^{-7} \times 350 = 0.7$ T.

7. Magnetisation Curves

We said previously that a core material which has a high permeability will produce a strong magnetic field in response to a smaller m.m.f excitation current. A set of magnetisation curves can be produced which shows the relationship between flux density, (B) and magnetising force, (H) for different core materials.

For ferromagnetic materials such as iron or steel, the relation between B and H is not linear as μ_r varies with flux density. Then magnetisation curves, or more commonly known as B-H Curves can be plotted for each type of core material used as shown in Figure 6.

Figure 6. Magnetisation of B-H Curves



The example set of magnetisation or B-H curves shown in Figure 6. represents the theoretical relationship between B and H for cast-iron and steel cores compared to air. Note that every type of core material will have its own set of magnetic hysteresis curves.

Flux density increases in proportion to the magnetic field strength until it reaches a certain value where it cannot increase any more becoming almost level and constant as the field strength continues to increase. This is because there is a limit to the amount of flux density that can be generated by the core as all the domains in the iron are perfectly aligned. So any additional increase will have no effect.

The point on the graph where the flux density reaches its limit is called **Magnetic Saturation** also known as **Saturation of the Core**. In our simple B-H curve example above the saturation point of the steel curve begins slightly above the 3200 ampere-turns per metre (3.2k AT/m).

Magnetic saturation occurs because the random haphazard arrangement of the magnetic domains (the atoms that behave as tiny magnets) within the core material change and become “lined-up” until they reach perfect alignment producing maximum flux density. So any increase in the magnetic field strength due to an increase in the electrical current flowing through the coil will have little or no effect as they cannot be aligned any more.

While the B-H curve above shows the core material becoming magnetised and saturated, an equal but opposite B-H curve would also exist for demagnetising the core. Then a complete B-H curve can be created to show the hysteresis loop of a particular magnetic core material.

8. Magnetic Hysteresis Curves

When a magnetic core, or material, is subjected to a cycle of magnetisation and demagnetisation, we would expect the ferromagnetic core material to reach its saturation point, maximum flux density, and then reduce to zero when H = 0. However, the magnetic flux does not disappear completely as the electromagnetic core material still retains some of its magnetism even when the current has stopped flowing through the coil.

The ability for an electromagnetic coil to retain some of its magnetism within the core after the magnetisation process has stopped is called **Retentivity**. The amount of flux density still remaining in the core is called Residual Magnetism or Residual Flux Density, (B_r). Where B is flux density and the subscript "r" stands for residual.

The reason for this retentivity is that some of the tiny molecular magnets of the core material do not return again completely random, as some remain aligned in the direction of the original magnetising field, thus giving them a sort of "memory".

Some ferromagnetic materials have a high retentivity (magnetically hard) making them excellent for producing permanent magnets. While other ferromagnetic materials have low retentivity (magnetically soft) making them ideal for use in electromagnets, solenoids or relays once the electrical power is removed.

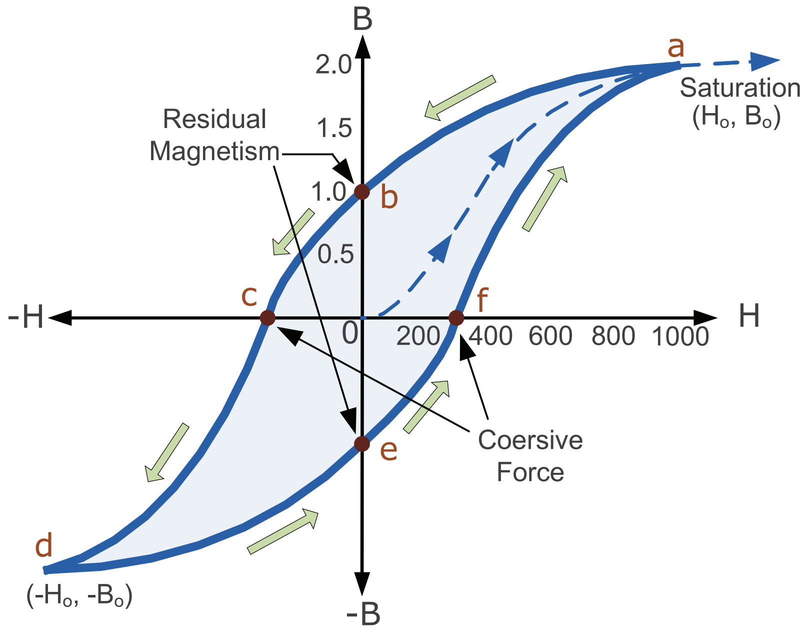
Reducing the residual flux density to zero is achieved by reversing the direction of current flowing through the coil, thus making the value of H negative. This effect is called a Coercive Force, (H_C). Increasing the reverse current will cause the core to reach its saturation point again, but in the reverse direction. Reducing the magnetising current again to zero will produce a similar amount of residual magnetism but in the reverse direction.

So by constantly changing the direction of the magnetising current through the electromagnetic coil from a positive direction to a negative direction, as would be the case in an AC supply, a Magnetic Hysteresis loop of the ferromagnetic core can be produced as shown in Figure 7.

The **Magnetic Hysteresis** loop of Figure 7 shows the behaviour of a ferromagnetic core graphically that the relationship between B and H is non-linear. Starting with a completely demagnetised core both B and H will be at zero, point **0** on the magnetisation curve.

As the magnetisation current is increased in a positive direction, the magnetic field strength H increases linearly so the flux density B also increases as shown by the curve from point 0 to point **a**, as it heads towards saturation.

Figure 7. A Typical Magnetic Hysteresis Loop



If the magnetising current in the coil is now reduced to zero, the magnetic field circulating around the core also reduces to zero. However, the coils magnetic flux will not reach zero due to the residual magnetism (retentivity) present within the core and this is shown on the hysteresis curve from point **a** to point **b**.

To reduce the flux density at point b to zero we need to reverse the current flowing through the coil. The magnetising force which must be applied to null the residual flux density called a "*Coercive Force*". This coercive force reverses the magnetic field rearranging the molecular magnets until the core becomes demagnetised at point **c**.

An increase in this reverse current causes the core to be magnetised in the opposite direction and increasing this magnetisation current further will cause the core to reach its saturation point but in the opposite direction, point **d** on the curve. Point **d** is symmetrical to point **a**. If the magnetising current is again reduced to zero, the residual magnetism present in the core will be equal to the previous value but in reverse at point **e**.

Reversing the magnetising current flowing through the coil this time into a positive direction will cause the magnetic flux to reach zero, point f on the curve. As before increasing the magnetisation current further in a positive direction will cause the core to reach saturation again and return back to point **a**.

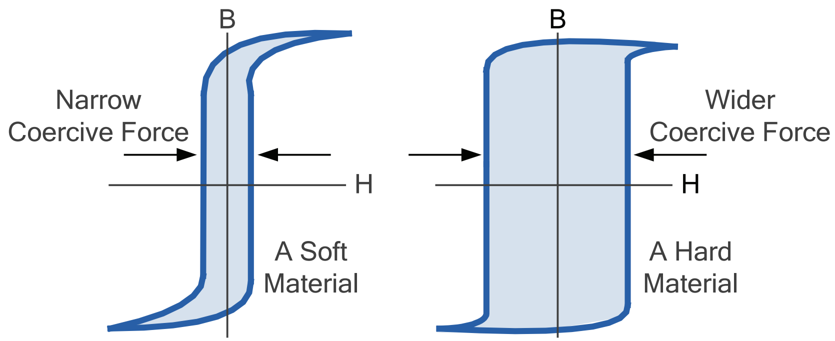
So as the magnetising current flowing through the coil alternates between a positive and negative value such as the cycle of an AC voltage, the B-H curve follows the complete path of **a**-**b**-**c**-**d**-**e**-**f** and back to **a**. This closed loop path is called a Magnetic Hysteresis Loop.

9. B-H Curves of Magnetic Materials

The effect of magnetic hysteresis shows that the magnetisation process of a ferromagnetic core can give the core a form of "memory" because it remains partially magnetised after the external magnetic field has been removed. Clearly for permanent magnets, high retenivity and coercivity is required to resist demagnetisation with the product of B_rH_c a desired parameter.

However, magnetically soft materials such as iron or silicon steel have very narrow B-H hysteresis loops and as such possess very small amounts of residual magnetism when de-energised. This makes these materials ideal for use in relays, solenoids and transformers as they can be easily magnetised and demagnetised.

Figure 8. Magnetic Hysteresis Loops for Soft and Hard Materials



By adding various additives to iron and steel such as silicon or cobalt, materials with a very wide coercive force can be made that have a large hysteresis loop. Materials with large hysteresis loops retain a high flux density and are therefore difficult to demagnetise.

> *Coercive Force is the magnetising force needed to reduce any residual magnetism to zero*

Since a coercive force must be applied to overcome this residual magnetism, work must be done in closing the hysteresis loop. However, the magnetic domains (molecular magnets) of the material resist being turned first in one direction and then in the other by the application of an electric current.

This results in a form of molecular friction in the core with the energy used to magnetise and demagnetise the core material being dissipated as heat. This heat is known as hysteresis loss, with the amount of loss depends on the material's value of coercive force.

Transformers and electrical machines that create an alternating flux are generally constructed from thin silicon steel sheets which are laminated together to produce larger magnetic cores. The lamination of magnetic cores greatly reduces hysteresis loss as the coercive force becomes extremely narrow reducing eddy-current losses in the core.

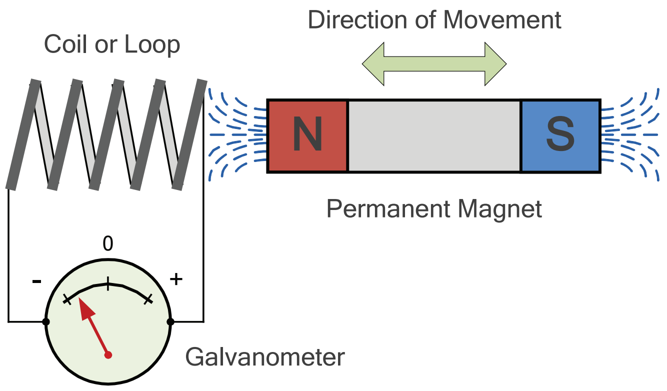
10. Electromagnetic Induction

We saw previously that when an electric current flows through a length of wire or conductor a magnetic field is created around it, and if the electric current changes value, so too does the strength of the magnetic flux.

But the same is also true in that if we pass or move a conductor or wire through a permanent magnetic field and emf will be induced into the wire. That is, an emf is produced due to the effect of induction by motion.

If we formed the length of wire into a closed circuit, then the induced emf (voltage) created by its movement through a magnetic field would cause a current to flow. This process is commonly known as **Electromagnetic Induction** and is the basic principle of operation of transformers, motors and generators. Consider Figure 9.

Figure 9. Electromagnetic Induction



When the permanent magnet of Figure 9. is moved "towards" the coil, the pointer or needle of the Galvanometer, which is basically a very sensitive centre zero moving-coil ammeter, will deflect away from its centre position in one direction only. When the magnet stops moving and is held stationary with regards to the coil the needle of the galvanometer returns back to zero as there is no physical movement of the magnetic field.

Likewise, when the magnet is moved "away" from the coil in the other direction, the needle of the galvanometer deflects in the opposite direction with regards to the first indicating a change in polarity. Then by moving the magnet back and forth towards the coil the needle of the galvanometer will deflect left or right, positive or negative, relative to the directional movement of the magnet field.

Now if the permanent magnet is held stationary and ONLY the coil is moved towards and away from the magnet, the needle of the galvanometer will also deflect in either direction. Then the action of moving a coil or loop of wire through a magnetic field induces a voltage in the coil with the magnitude of this induced voltage being proportional to the speed or velocity of the actual movement.

So the faster the relative motion between the coil and the magnetic field, the greater will be the amount of induced emf in the coil. In other words, there must be "relative motion" or movement between the coil and the magnetic field as either the magnetic field, the coil, or both can move.

11. Faraday's Law of Induction

So a distinct relationship exists between an electrical voltage and a changing magnetic field to which Michael Faraday's famous law of electromagnetic induction states: "*that a voltage is induced in a circuit whenever relative motion exists between a conductor and a magnetic field and that the magnitude of this voltage is proportional to the rate of change of the flux*". So how much voltage (emf) can be induced into a coil or wire using just magnetism.

Electromagnetic induction depends upon the following three different factors:

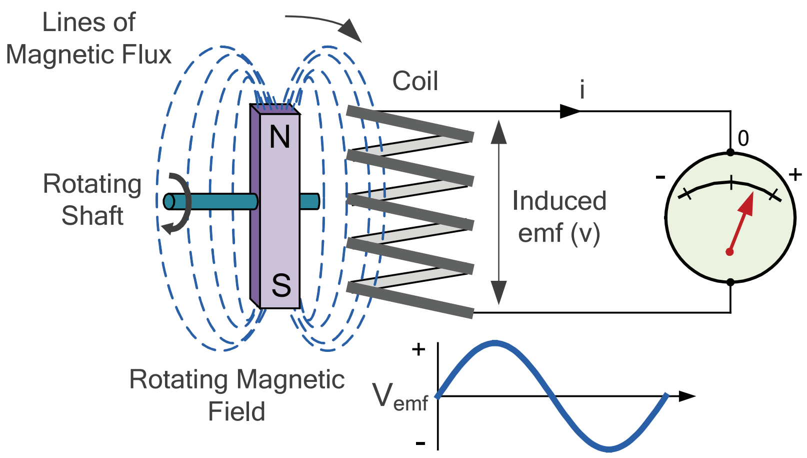
1. Increasing the number of turns of wire in the coil – By increasing the amount of individual conductors cutting through the magnetic field, the amount of induced emf produced will be the sum of all the individual loops of the coil, so if there are 20 turns in the coil there will be 20 times more induced emf than in one piece of wire.
2. Increasing the speed of the relative motion between the coil and the magnet – If the same coil of wire passed through the same magnetic field but its speed or velocity is increased, the wire will cut the lines of flux at a faster rate so more induced emf would be produced.
3. Increasing the strength of the magnetic field – If the same coil of wire is moved at the same speed through a stronger magnetic field, there will be more emf produced because there are more lines of force to cut.

If we were able to move the permanent magnet above in and out of the coil at a constant speed and distance without stopping, we would generate a continuously induced voltage.

This voltage would continuously alternate between one positive polarity and a negative polarity producing an alternating or AC output voltage. This is the basic principle of how an electrical generator works.

In small generators such as a bicycle dynamo, a small permanent magnet is rotated by the action of the bicycle wheel inside a fixed coil as shown in Figure 10.

Figure 10. A Simple Generator using Magnetic Induction



The simple dynamo type generator above consists of a permanent magnet rotating around a central shaft with a coil of wire placed next to this rotating magnetic field. As the magnet spins on its axis, the magnetic field around the top and bottom of the coil constantly changes between a north and a south pole. The fields rotational movement results in an alternating emf being induced into the coil as defined by Faraday's law of electromagnetic induction.

Thus, if a conductor of length, ℓ metres moves at right angles to a uniform magnetic field of **B** in Wb/m^2 at a rotational velocity of **v** in meters/second or m/s, through a distance dx in dt seconds. Then the magnitude of the emf induced in a conductor or coil which is directly proportional to the rate of change of flux linkage is giving by the motional emf expression of:

$$\varepsilon_{induced} = -N\frac{d\Phi}{dt} = -\frac{B\ell dx}{dt}$$

$$\therefore\ \varepsilon_{induced} = -B\ell v \quad \text{volts}$$

If the conductor does not move at right angles (90°) to the magnetic field, then the angle θ° should be added to the above expression giving a reduced output voltage as the phase angle increases:

$$\varepsilon_{induced} = -B\,\ell\, v\,\sin\theta \quad \text{volts}$$

So if a conductor of length 2m, is moving at through a magnetic field of 0.7 T at a speed of 5 m/s, then the induced voltage ε would be: **ε = -B ℓ v = 0.7 × 2.0 × 5.0 = -7.0 volts**.

12. Lenz's Law of Electromagnetic Induction

Faraday's Law tells us that inducing a voltage into a conductor or coil can be done by either passing it through a magnetic field, or by moving the magnetic field past it. A voltage known as an induced emf will be induced into the conductor (or coil) by the changing magnetic field.

But this induced emf causes a current to flow through the coil which itself will produce its own magnetic field. This additional magnetic field creates a secondary emf which opposes the change that is causing it and the faster the rate of change of current the greater is the opposing emf. This self-induced emf will, by Lenz's law oppose the change in current in the coil and because of its direction this self-induced emf is commonly called a back-emf.

Lenz's Law states that "*the direction of an induced emf is such that it will always opposes the change that is causing it*". In other words, an induced current will always OPPOSE the motion or change which started the induced current in the first place. Likewise, if the magnetic flux is decreasing then the induced emf will oppose this decrease by generating an induced magnetic flux that aids or reinforces the original flux.

Lenz's law is one of the basic laws in electromagnetic induction for determining the direction of flow of induced currents and is related to the law of conservation of energy. According to the law of conservation of energy which states that the total amount of energy in the universe will always remain constant as energy cannot be created nor destroyed. Lenz's law is derived from Michael Faraday's law of induction.

13. Self-Inductance

Having said previously that a changing electric current in a coil of wire induces an electromotive force (emf) within that same coil which opposes the change. If the coil creates a magnetic flux in its core which links all the N turns of the coil, as the flux changes with time (caused by current varying with time) the induced emf in the coil produces or creates a back-emf (or counter-emf). The property of the coil to oppose any change in the amount of current flowing through it is called **Self-inductance** or simply **Inductance**.

This property of self-inductance does not prevent the current flowing through the coil from changing, what it does is slow down or restrict the actual change. So the greater the back-emf voltage, the greater the self-inductance of the coil and therefore, the greater is the opposition to the changing current.

Now according to Faraday's laws of electromagnetic induction above, the magnitude of the induced voltage in a coil depends upon the number of coil turns (N) and the di/dt rate of change of current with time (usually given in amperes per second, or amp/sec).

$$\varepsilon_{induced} = -N\frac{d\Phi}{dt} \times \frac{di}{dt} \quad \text{volts} \quad \text{(as per Lenz's Law)}$$

Since the flux is due to the current in the coil, it follows that flux linkages (NΦ) will be proportional to **i**. Thus:

$$\therefore \varepsilon_{induced} = -L\frac{di}{dt} \quad \text{volts}$$

Where L is the constant of self-inductance of the coil. Note that the standard unit of self-inductance is the henry (H). Then if the induced voltage, ε = 1 volt, and di/dt is changing at a rate of 1 Ampere/second, self-inductance L would be exactly equal to 1 Henry.

Note that the minus sign represents Lenz's law mathematically because the current flows in the opposite direction to oppose the change. Also, this general expression for the self-inductance (L) of a coil can usually be found at any point on the B-H curve of the magnetic core material used.

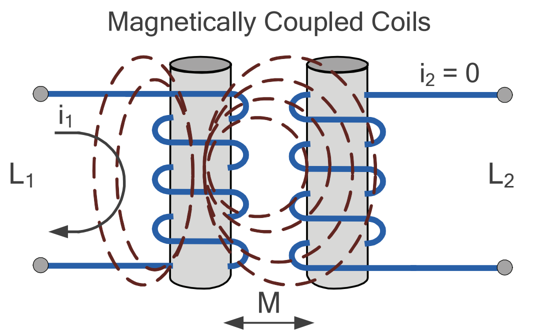
14. Mutual-Inductance

Thus far we have considered one single coil on its own, but when two (or more) coils are wound together onto a common core or placed in close proximity to each other, part of the flux produced by one coil will also link to the other coil. So we can correctly say that the two coils are magnetically coupled together.

The magnetic coupling between two coils leads to the concept of mutual inductance. As we have seen previously, each coil will have its own value of inductance, (L) but due to the voltage produced by the coupling effect between the two coils, additional inductance is added to each coil.

Then the two inductively coupled coils will have mutual inductance because the change of current in one coil induces a voltage in the other. The unit given for mutual inductance is the henry (H). Consider the two magnetically coupled coils shown in Figure 11.

Figure 11. Two magnetically coupled coils



Here the current flowing in coil one (L_1), sets up a magnetic field around itself with some of its magnetic lines of flux passing through and into coil two (L_2). Thus giving us mutual inductance. Coil one has a current of I_1 and N_1 turns, while coil two has N_2 turns.

Thus, the mutual inductance of coil two that exists with regards to coil one, $M_{1\text{-to-}2}$ depends on their physical positions with respect to each other. Then the mutual inductance that exists between the two coils is given as:

$$M_{1\text{-to-}2} = \frac{N_2 \Phi_{1\to 2}}{I_1}$$

Likewise, the flux linking coil one when a current flows around coil two is exactly the same as the flux linking coil two when the same current flows around coil one. Then the mutual inductance of coil one with respect of coil two is defined as $M_{2\text{-to-}1}$. This mutual inductance is true irrespective of the size, number of turns, relative position or orientation of the two coils. Then we can write the mutual inductance between the two coils as: $M_{1\rightarrow 2} = M_{2\rightarrow 1} = M$.

For two magnetically coupled coils, their self-inductances will be: L_1 and L_2. So the voltage induced into coil one would be: $v_1 = L_1(di_1/dt)$ and the voltage induced into coil two would be: $v_2 = L_2(di_2/dt)$.

Thus the mutually induced voltage between the two coils would be: $v = M(di/dt)$. Where M corresponds to their mutual inductance as we have seen previously.

15. The Coefficient of Coupling

The *coefficient of coupling*, (*k*) that exists between two adjacent coils is defined as being the fractional amount of magnetic flux created by the current carrying coil that links magnetically to the other coil. Generally, the amount of inductive coupling that exists between the two coils is expressed as a fractional number between 0 and 1 instead of a percentage (%) value, where 0 indicates zero (0%) or no inductive coupling, and 1 indicating full (100%) or maximum inductive coupling.

In other words, if **k = 1** (i.e. 100%) the two coils are perfectly coupled, if k is greater than 50% (0.5) the two coils are said to be tightly coupled and if k is less than 50% the two coils are said to be loosely coupled. So if the two coils each have self-inductances of L_1 and L_2, then the mathematical expression given for the mutual inductance M that exists between them is given as:

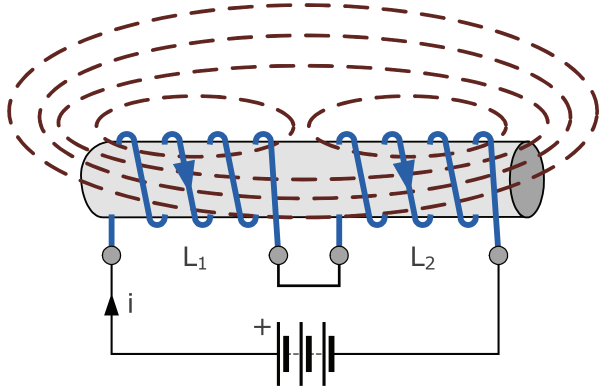
$$M = k\sqrt{L_1 L_2} \quad \therefore\ k = \frac{M}{\sqrt{L_1 L_2}}$$

When the coefficient of coupling, k is equal to 1, (unity) such that all the lines of flux of one coil cuts all of the turns of the second coil, that is the two coils are tightly coupled together, for example, one coil wound on top of the other.

The resulting mutual inductance will be equal to the geometric mean of the two individual inductances of the coils. Also, the larger the coefficient of coupling, **k** the larger will be their mutual inductance, **M**.

We have seen that when two (or more) coils are brought close together, the magnetic field of one links with the other with the effect of mutual inductance either increasing or decreasing the total inductance depending upon the amount of magnetic coupling, **k**. But let us also consider the total inductance of series coils wound onto the same magnetic core as shown in Figure 12.

Figure 12. Coupled Series Coils



If two coils (or inductors) are connected together in series and have mutual coupling between them, their effective coil inductances will no longer simply be L_1 and L_2. Since the current is the same for both coils (series circuit), the induced voltage of the two coils will therefore be:

$$v_1 = L_1\frac{di}{dt} \pm M\frac{di}{dt} = [L_1 \pm M]\frac{di}{dt}$$

therefore, coil 1 has an effective inductance value of: $L_1 \pm M$

and:

$$v_2 = L_2\frac{di}{dt} \pm M\frac{di}{dt} = [L_2 \pm M]\frac{di}{dt}$$

therefore, coil 2 has an effective inductance value of: $L_2 \pm M$

Thus the total inductance, L_T is given as:

$$L_T = L_1 + L_2 \pm 2M \quad \text{henries (H)}$$

Where: ± 2M denotes *cumulatively coupled* or *differentially coupled* coils.

End of the Electromagnetism eBook

Last revision: June 2022

With the completion of this Magnetism and Electromagnetism eBook you should have gained a basic understanding and knowledge of magnetic circuits and coils. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 10 we will learn more about electromagnetism and how it is used in Transformers.

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